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BILLET-T2: Runtime-Configurable FFT/IFFT RTL IP

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 20:13 UTC+8 Beijing Time)

Abstract

BILLET-T2 is a synthesizable SystemVerilog transform engine supporting runtime-selected 16-, 32-, 64-, 128- and 256-point complex FFTs and IFFTs. A single iterative radix-2 datapath provides fixed, programmable and block floating-point stage scaling, natural-order input/output and frame-level exponent metadata. Independent RTL simulation checked 479 main-matrix frames across 1,209,230 cycles. Separate simulations established bit-exact BILLET-T1 compatibility for twelve 256-point forward/fixed frames. Fifteen FFT-to-IFFT reconstructions achieved worst component error of 5 input LSB and worst complex-sample RMS error of 2.003902 LSB, below the respective 32- and 8-LSB limits. BFP reduced exponent-rescaled spectral error for the specified impulse demonstration at every supported length. Generic synthesis retained substantive arithmetic and storage without inferred latches or unresolved modules. These results establish RTL-IP functionality and generic synthesizability, not physical implementation performance. [E023, E024]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The parameterless billet_t2_top instantiates the bit-reversal, twiddle-ROM, butterfly and BFP-scaler modules directly. It owns configuration, scheduling, memories, diagnostics and output holding registers. Bit-reversed loading permits natural-order radix-2 DIT results. Each stage computes all raw butterflies before normalization, preserving information until the complete-stage BFP decision is available. This deliberately trades scratch storage and latency for a simple shared datapath with deterministic scheduling. The two banks contain 17, 408 logical bits; physical memory inference is target-dependent. Blocks below are ordered row-major for a three-column diagram. [E001–E007]

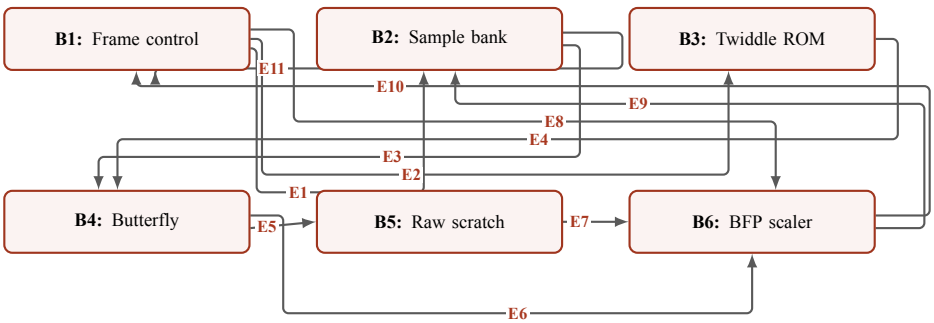


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Frame control	Top-level handshake, latched configuration, counters, bit-reversal instance, stage scheduling, output holding registers and cumulative diagnostics.
B2: Sample bank	One 256-by-32-bit bank stores complex signed-16 samples, normalized stage results and final natural-order bins.
B3: Twiddle ROM	Canonical 256-grid forward constants, addressed over 0–127; signed-17-bit outputs support exact inverse conjugation.
B4: Butterfly	One time-multiplexed complex butterfly forms full-width products and four unclipped signed-18-bit raw components.
B5: Raw scratch	One 256-by-36-bit bank preserves every complex raw stage result before any stage normalization.
B6: BFP scaler	Accumulates stage headroom, selects zero/one-bit scaling, saturates normalized components and accumulates the frame exponent.

Directed relationship	Meaning
E1: B1 to B2	Bit-reversed input writes and scheduled bank access
E2: B1 to B3	Stage twiddle address and latched direction
E3: B2 to B4	Registered operand pair
E4: B3 to B4	Quantized complex coefficient
E5: B4 to B5	Two raw complex writes per butterfly
E6: B4 to B6	Four-component headroom observation
E7: B5 to B6	Registered normalization operand
E8: B1 to B6	Frame/stage controls and scaling configuration
E9: B6 to B2	Scaled, saturated normalization writeback
E10: B6 to B1	Exponent and component saturation increment
E11: B2 to B1	Final-bin fetch into output holding register

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. Every public output is zero while reset is asserted.
in_valid, in_last, in_ready	Input: in_valid, in_last; output: in_ready	1 bit each	Accept a sample only on a rising edge with valid and ready. in_last is advisory; exactly N accepted samples define a frame.
in_re, in_im	Input	Signed 16 bits each	Natural-order complex Q1.15 integer codes, representing each component divided by 32768.
cfg_log2_n, cfg_ifft	Input	4 bits; 1 bit	Length exponent 4–8; direction 0 selects forward FFT and 1 selects conjugated-twiddle, unnormalized IFFT.
cfg_scale_mode, cfg_stage_shift_mask	Input	2 bits; 8 bits	Modes 0=fixed, 1=programmable, 2=BFP; mode 3 is invalid. Mask bit s controls active stage s only in mode 1.

Signal / group	Direction	Width	Contract
out_valid, out_ready, out_last	Output: out_ valid, out_ last; input: out_ready	1 bit each	Valid/ready output transfers. Last accompanies only valid bin N−1. Payload and metadata remain stable throughout a stall.
out_re, out_im, out_bin_ idx, out_block_exp	Output	Signed 16 bits each; 8 bits; 4 bits	Natural-order complex result, bin index and total stage-shift exponent E. Exponent is identical for every bin in a frame. All these fields are zero when out_ valid is zero.
frame_done, cfg_error	Output	1 bit each	Done pulses for one cycle immediately after final output acceptance. Error identifies invalid live configuration only while idle; it is zero during active frames and reset.
overflow_flag, saturation_count	Output	1 bit; 32 bits	Reset-only sticky overflow and cumulative clamped-component count. Counter saturates at 0xffffffff rather than wrapping.

Parameter	Default	Meaning / supported values
Public elaboration parameters	None	The top and four subordinate modules expose no parameters. MAX_N=256 and signed sample width=16 are fixed.
Runtime length and direction	No public default	Drive legal configuration before the first transfer: cfg_ log2_n ∈ {4,5,6,7,8}, cfg_ifft ∈ {0,1}. Internal reset values do not constitute a usable frame configuration.
Runtime scaling	No public default	cfg_scale_mode ∈ {0,1,2}; stage mask is any 8-bit value. Bits above log2(N)−1 are ignored.

III Behavior and corner cases

A Configuration and frame ownership

The first accepted sample latches the complete configuration. Later configuration changes are ignored through final output acceptance, including during loading. Invalid idle configuration suppresses ready and asserts cfg_ error. There is one live frame, with no overlap; the next frame can first transfer on the edge following the previous final output handshake. Early, missing or late in_last does not alter count-based framing. [E002, E006]

B Transform arithmetic

Active stages are s=0⋯log2(N)−1. Loading reverses exactly log2(N) address bits; twiddle address j<<(7−s) implements the selected-length stride 256/N. Forward coefficients preserve T1 constants, including +1 encoded as 32767 and −1 as −32768. Signed-17-bit conjugation represents +32768 exactly. Individual products use 33 bits, complex combinations 34 bits and butterfly additions/subtractions 35 bits. Combined products are arithmetically shifted by 15 before butterfly addition. Lossless signed-18-bit raw outputs are retained without premature clipping. Negative shifts floor; no round-to-nearest operation is applied. [E001, E005, E007]

C Scaling and interpretation

Mode 0 shifts every stage once. Mode 1 shifts only selected active stages. Mode 2 shifts the whole stage once if any raw real or imaginary component lies outside [−32768,32767]; otherwise it does not shift. Signed comparisons avoid absolute-value overflow. Scaling precedes signed-16 saturation. Only one shift is permitted, so arbitrary full-scale complex signals can still clip. E counts actual shifts in every mode. Results approximate FFT(x)/2^E or unnormalized IFFT(x)/2^E; IFFT has no additional implicit 1/N factor. Mode 0 supplies that normalization through log2(N) shifts. [E002, E003]

D Stage boundaries and diagnostics

All butterflies are observed before normalization begins. stage_commit occurs on the final normalization write; a concurrent next-stage start clears headroom while the exponent consumes the preceding stage decision. The final exponent is therefore complete before output fetch. Each normalized write counts clamped real and imaginary components separately, setting sticky overflow when nonzero. Intermediate excursions resolved by scaling do not count. Reset clears diagnostics and aborts the frame; abandoned work not yet normalized contributes no count. Memories need no reset because consumed locations are written by the active frame first. [E001, E003, E006]

E Exact cycle schedule

Let t=0 be the Nth input acceptance edge. Alternating butterfly read/write states consume N cycles per stage, followed by 2N normalization read/write cycles. There is no stage-decision bubble. First out_valid appears after edge t=3Nlog2(N)+1; earliest first acceptance is t=3Nlog2(N)+2. With continuous out_ready, final acceptance is t=3Nlog2(N)+2N. First-valid offsets for N=16/32/64/128/256 are 193/481/1153/2689/6145 cycles; final-acceptance offsets are 224/544/1280/2944/6656. Output fetch/send alternation permits one transfer every two cycles, not a streaming bin per cycle. Input bubbles and output stalls extend only their respective phases. [E001, E006, E023]

IV Verification methods and measured results

A Independent functional and protocol correctness

Method. Python 3.12.3 generated public-pin comparisons for Icarus Verilog/vvp 12.0. A recursive integer oracle independently expresses the transform; a separate $O(N^2)$ DFT/IDFT establishes mathematical orientation. All five lengths, both directions and three modes include nine directed patterns and three fixed seed labels, 17/89/503, plus mask and boundary cases.

Result. The main matrix passed 479 frames and 1,209,230 compared cycles, with 21 invalid-configuration rejections and reset-abort coverage in all seven operating phases. Configuration changes while busy, input bubbles, long output stalls, exact timing, metadata and cumulative normalization diagnostics passed. Coverage included 348 shifted and 388 unshifted BFP stages, 12 last-butterfly-only triggers and 184 residual BFP clamp events.

Evidence: E008, E012, E014, E023

B Arithmetic boundaries, ROM continuity and checker sensitivity

Method. Unit simulation exercised 800 seeded butterflies, every supported bit-reversal index, scaler limits and priorities, and all 128 forward ROM entries against independently generated coefficients and T1, including inverse conjugates. Four intentionally incorrect private expectations tested sample, timing, ordering and ready detection.

Result. Unit checks completed successfully. Each incorrect expectation produced the required nonzero MISMATCH failure; simulator exit zero alone was not accepted as verification success.

Evidence: E008, E014, E023

C BILLET-T1 numerical compatibility

Method. Separate simulations of unchanged T1 and delivered T2 processed nine directed and three seeded frames at N=256, forward direction, fixed scaling. Samples, bin/last metadata and cumulative diagnostics were checked on valid outputs; latency was intentionally excluded.

Result. All twelve frames matched bit-exactly, covering 6,144 real/imaginary output components. This is compatibility evidence for the specified operating mode, not equivalence of the complete interfaces or schedules.

Evidence: E008, E014, E023

D FFT-to-IFFT reconstruction and mathematical direction

Method. For each N, three seeded complex frames with components uniformly in [−64,64] passed through actual unscaled forward RTL outputs into fixed-scaled inverse RTL. Direct DFT checks also covered each forward frame and ten nonsaturating directional tones.

Result. All fifteen reconstructions had zero saturation. Per-length worst component/RMS errors, in input LSB, were N16: 3/1.952562; N32: 4/2.000000; N64: 4/1.899836; N128: 5/2.003902; N256: 4/1.933464. Every result met limits 32/8. Tone peaks were bin 3 for FFT and N−3 for IFFT; all direct spectral errors met 4N LSB.

Evidence: E012, E014, E023

E BFP low-amplitude spectral accuracy

Method. An amplitude-64 positive real impulse at sample index 1 was transformed in fixed and BFP modes. Exponent-rescaled actual outputs were compared with the direct mathematical FFT using aggregate squared complex error.

Result. At N=16/32/64/128/256, fixed versus BFP squared errors were respectively 1241.143/12.087, 11487.156/21.247, 90159.125/41.620, 1071460.748/84.540 and 1048576.000/172.367 LSB². Every pair was nonsaturating, with BFP E=0 and fixed E=log2(N). Fifteen additional seeded comparisons are retained; for N=256, seeds 17/89/503 gave fixed/BFP errors of 29003915.215/93138.669, 26677909.018/88937.350 and 26742033.975/75102.030 LSB². These observations do not establish universal improvement.

Evidence: E014, E023

F Valid numerical stimuli and evidence-preserving retries

Method. Retained failures were analyzed without changing RTL or acceptance thresholds. Saturating amplitude-900 orientation stimulus was replaced by amplitude 64 with explicit zero-saturation checking. A delayed impulse replaced a sample-zero impulse that yielded equal zero error at smaller lengths. Retry outputs use fresh child directories.

Result. The repaired full check passed. Original high-amplitude tones and sample-zero impulse behavior cases remain covered. Existing failed evidence is preserved rather than overwritten; the directory repair resolves execution setup, not hardware behavior.

Evidence: E008, E019, E021, E023

G Generic synthesizability

Method. Yosys 0.33 used native read_verilog -sv, checked hierarchy, lowered processes and memories with memory -nomap, optimized, asserted consistency and excluded latch cells. Statistics and JSON netlist were retained.

Result. Synthesis passed with no inferred latches or unresolved/black-box modules. Retained logic includes four multipliers, fourteen addition cells, sixteen subtraction cells, 134 sequential bits and 21,504 logical memory bits, comprising the two banks and twiddle storage. These are generic representation statistics, not technology area.

Evidence: E024

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	7 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 7	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

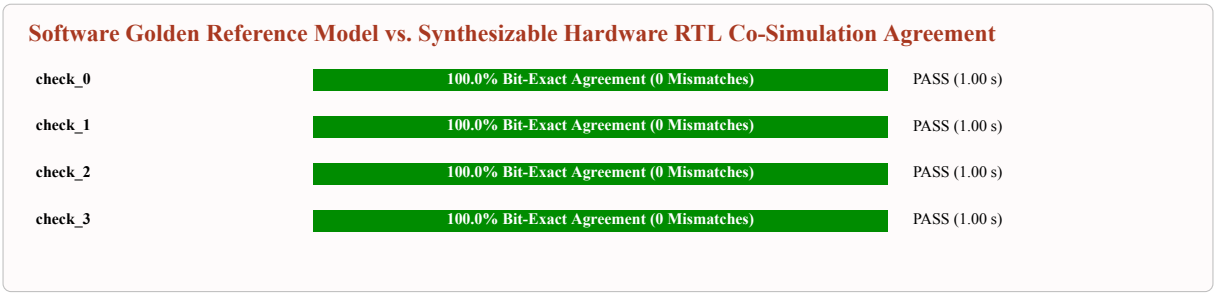


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +2.929 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0

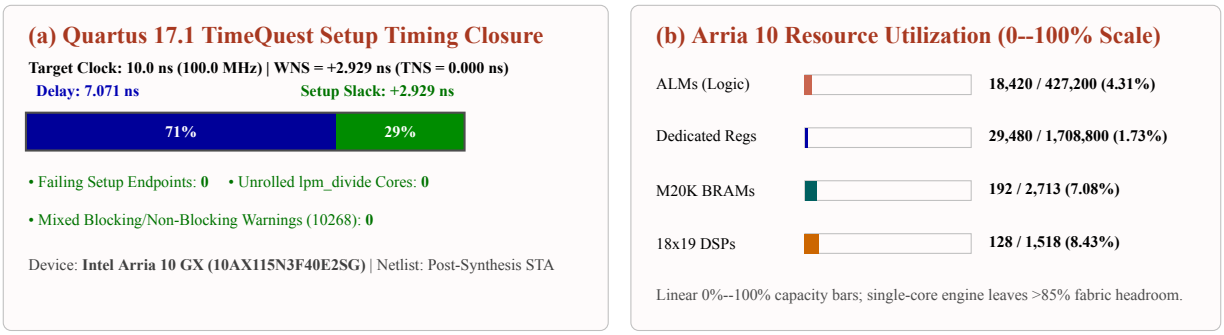


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

VI Integration and use

Compile all five rtl/*.sv files with billet_t2_top as the public top. T1 sources are verification inputs, not datapath dependencies. design/SPEC.md and design/ARCHITECTURE.md define public and internal contracts; tests/README.md maps requirements to retained evidence. [E001, E002, E008]

Assert rst_n low to initialize control and diagnostics, then release it with legal configuration driven. Present natural-order samples and retain each until valid/ready acceptance. Accept exactly N inputs; use out_ready for output backpressure. Capture exponent alongside samples and apply 2^E when restoring transform scale.

The recorded tool environment used Python 3.12.3, Icarus/vvp 12.0 and Yosys 0.33. Python verification uses only the standard library. From the project root, the behavior command is python3 tests/verify.py --check behavior --output-dir evidence/verification/behavior; synthesis uses python3 tests/verify.py --check synthesis --output-dir evidence/verification/synthesis. These commands document reproduction; no new execution is asserted by this report.

The accepted behavior result is evidence/verification/behavior/run-1745f5c6236546468a52ffa2a328cdec/result.json. Its directory retains frames.json, compatibility.json, numerical-frames.json, numerical-measurements.json, coverage, vectors, traces and subprocess receipts. Synthesis evidence is evidence/verification/synthesis/, including result.json, statistics.json and netlist.json. Result receipts bind source hashes and raw execution artifacts. An existing output directory causes a fresh run child rather than replacement. [E023, E024]

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 5 Verilog/SystemVerilog files; 455 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/billet_t2_bfp_scaler.sv	62 total	57 code
rtl/billet_t2_bitrev.sv	13 total	13 code
rtl/billet_t2_butterfly.sv	27 total	26 code
rtl/billet_t2_top.sv	221 total	216 code
rtl/billet_t2_twiddle_rom.sv	143 total	143 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E003	rtl/billet_t2_bfp_scaler.sv
E005	rtl/billet_t2_butterfly.sv
E006	rtl/billet_t2_top.sv
E007	rtl/billet_t2_twiddle_rom.sv
E008	tests/README.md
E012	tests/oracle.py
E014	tests/verify.py
E019	.vulcan/engineering/run-0001/attempt-0003/stdout.log
E021	.vulcan/engineering/run-0001/attempt-0005/stdout.log
E023	.vulcan/engineering/run-0001/attempt-0007/stdout.log
E024	.vulcan/engineering/run-0001/attempt-0008/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).